

MIKE JATTU

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Education

University of Alberta

BSc. Honors in Computer Science with AI Option

Grad: May 2027

Edmonton, Alberta

Publications

Oil-Sands Level Identification in a Dump Hopper

2025

University of Alberta | Manuscript in Preparation

Edmonton, Alberta

- **Jattu, S.K.**, Sekhon, K.H.S., Bhase, S.S., & Huang, B. – Developed a vision-based soft-sensing framework using DeepLabV3 semantic segmentation and contour-based post-processing to estimate oil-sands hopper levels from industrial camera footage, outperforming classical thresholding approaches in reliability and interpretability.

Work Experience

Undergraduate Research Assistant

Jan 2025 – Sept 2026

PDASA Lab, Department of Chemical Engineering, University of Alberta

Edmonton, Alberta

- **Research Project:** Investigated computer vision approaches for industrial monitoring in oil sands environments, contributing to applied ML research in safety-critical systems.
- **Impact:** Developed and trained an ML model to automate sand level monitoring in dump hoppers, achieving **92% accuracy** and reducing manual inspection requirements.
- **Supervisor:** Dr. Biao Huang

Data Scientist

May 2025 – Dec 2025

City of Edmonton | ( [Open Source Code](#))

Edmonton, Alberta

- **Project:** Developed and Deployed a vehicle detection ML model with an optimized data pipeline for continuous training and lifecycle management.
- **Impact:** Architected and deployed a real-time, open-source traffic control system achieving approximately **65% cost savings** in city infrastructure operations for vehicle detection.

Software Engineer Intern

May 2024 – Apr 2026

Ryan Tax Firm

Mississauga, Ontario (Remote)

- **Project:** Develop Model Context Protocol (MCP) based chat bots that enables utilization of multiple data sources efficiently.
- **Impact:** Streamlined the lead qualification process by building a web application; **reducing retrieval time by 60%**

Teaching Experience

Undergraduate Teaching Assistant

Sept 2024 – Present

Department of Mathematics and Statistics, Department of Computer Science

Edmonton, Alberta

- **CMPUT 174:** Introduction to the Foundations of Computation I
- **CMPUT 101:** Introduction to Computing
- **CMPUT 204:** Algorithms I
- **CMPUT 229:** Computer Organization and Architecture I
- **CMPUT 328:** Computer Vision
- **Mathematics Help Center:** Honors Calculus I/II/III, Honors Linear Algebra I/II

Summer Schools & Training

CVR Vision Science Summer School

June 8 – 12, 2026

Centre for Vision Research, York University

Toronto, Ontario

- Selected to attend a competitive, tuition-free intensive program hosted by the Centre for Vision Research, featuring lectures by faculty in human visual perception, computer vision, machine learning, and visual neuroscience.
- Program theme: *Vision Science in the Age of AI* – exploring how AI models serve as explicit, testable computational hypotheses about brain function, with hands-on tutorial projects and career development panels.

Presentations

Oil Sands Level Estimation via Computer Vision

Dec 2025 & Apr 2026

PDASA lab Bi-Annual Industry Meeting, University of Alberta

Edmonton, Alberta

- Presented the research on oil sands hopper level estimation pipeline.

Uncertainty Quantification in Segmentation-Based Level Estimation

2025

PDASA Lab, Department of Chemical Engineering, University of Alberta

Edmonton, Alberta

- Delivered a literature review presentation on Uncertainty Quantification for deep segmentation models.

Distribution-Free Uncertainty Quantification: Split Conformal Prediction

2025

Data Science Team, City of Edmonton

Edmonton, Alberta

- Presented a technical seminar on split conformal prediction for regression and classification.

Projects

- **Guardian:** AI-powered scenario-based learning platform helping ESL learners prepare for the Alberta Basic Security Training (ABST) exam with multilingual and voice support. ( [Github](#)) 2026
- **RideSafe:** Real-time AI surveillance app detecting threats on public transit using a Vision Transformer model with instant rider safety reporting. ( [Github](#)) 2025
- **FoodMinder:** Smart AI powered grocery management app using OCR receipt scanning, expiry reminders, recipe suggestions for near-expiry items, and nearby food donation center locating. ( [Github](#)) 2025

Awards & Honors

- 3rd Place in DevCon Hackathon 2026
- 1st Place in Hack4Health Hackathon 2025
- 2nd Place in DevCon Hackathon 2025
- Dean's Honor Roll 2022–2023 & 2024–2025
- University of Alberta Undergraduate Scholarship (International) 2023–2024
- Faculty of Science Gold Standard Scholarship 2022–2023
- University of Alberta Regional Excellence Scholarship 2022–2023

Leadership & Extracurriculars

Vice President of Administration

May 2025 – Dec 2025

University of Alberta Computing Science Club (UACS)

Edmonton, Alberta

- **Activities:** Founded and organized **Beyond Code**, a bi-weekly research seminar series connecting undergraduate CS students with graduate researchers; ran **10 sessions** covering topics including NLP, differential privacy, deep continual learning and others
- **Impact:** Helped various undergraduate Students learn about different research fields in Computing Science.